# RO2 - Lecture 6
## Complementary Filter

we focus on students

Fachhochschule Dortmund
University of Applied Sciences and Arts

## Summary for orientation estimation based on accelerometer/gyroscopes:

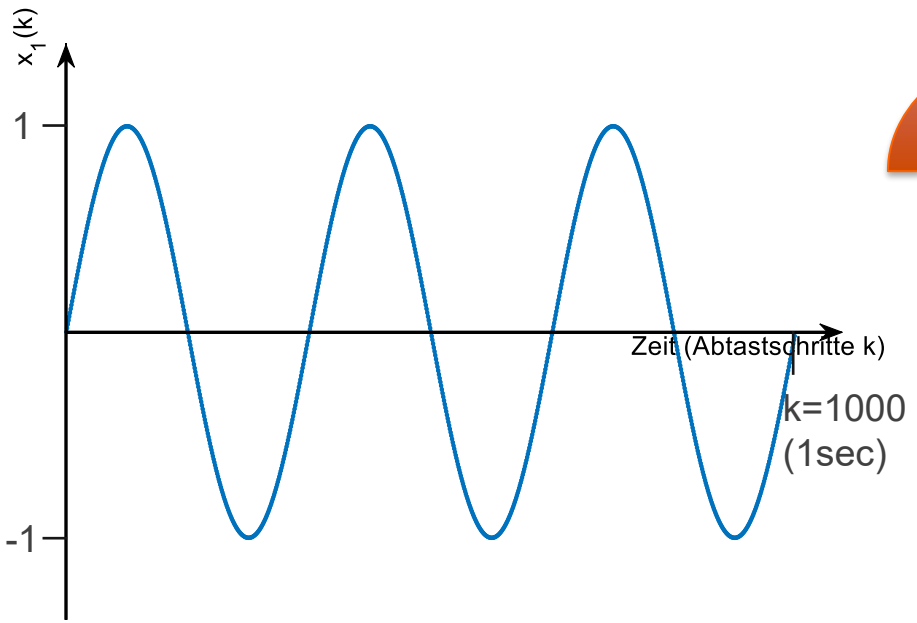
- Accelerometers provide good orientation estimation under static conditions, but the estimation becomes error-prone during dynamic motion because linear acceleration is indistinguishable from gravity
- Gyroscopes provide good orientation estimation under dynamic conditions, but the resulting orientation data drifts over time because the angular rate measurements must be integrated

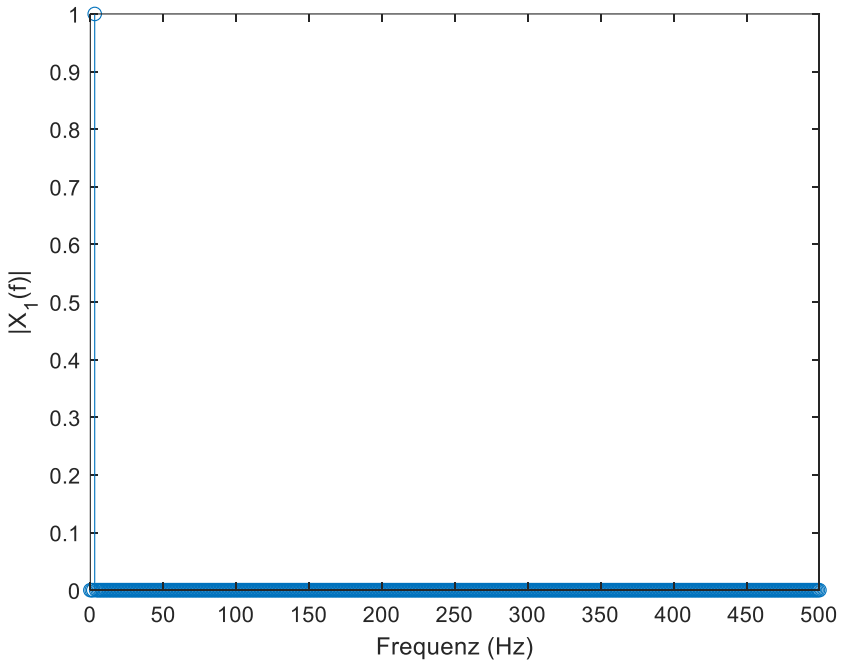
## Complementary Filter for Orientation Estimation

- Combines accelerometer and gyroscope data to leverage their strengths
- Implements a weighted fusion of the two data sources:
  - High-pass filtered gyroscope data (focuses on short-term changes)
  - Low-pass filtered accelerometer data (focuses on long-term orientation)

Recap on Digital Filters…

Fourier-Transformation
(DFT, FFT)



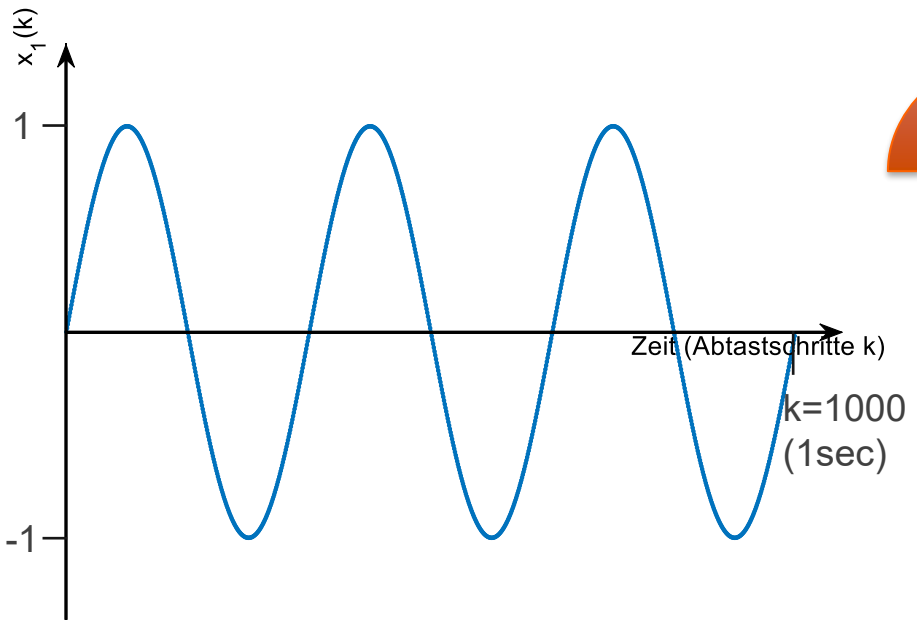



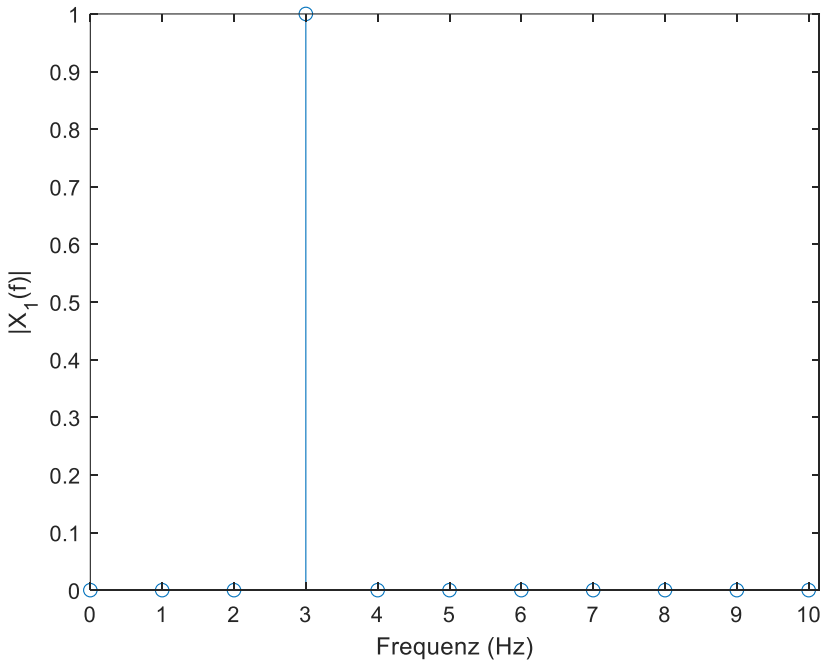

$$T_s = 1\text{ms}\,, F_s = \frac{1}{1\text{ms}} = 1\text{kHz}$$

$$x_1(\text{kT}) = \sin(2\pi F_1 kT)$$

$$F_1 = 3\text{Hz}$$

Fourier-Transformation
(DFT, FFT)

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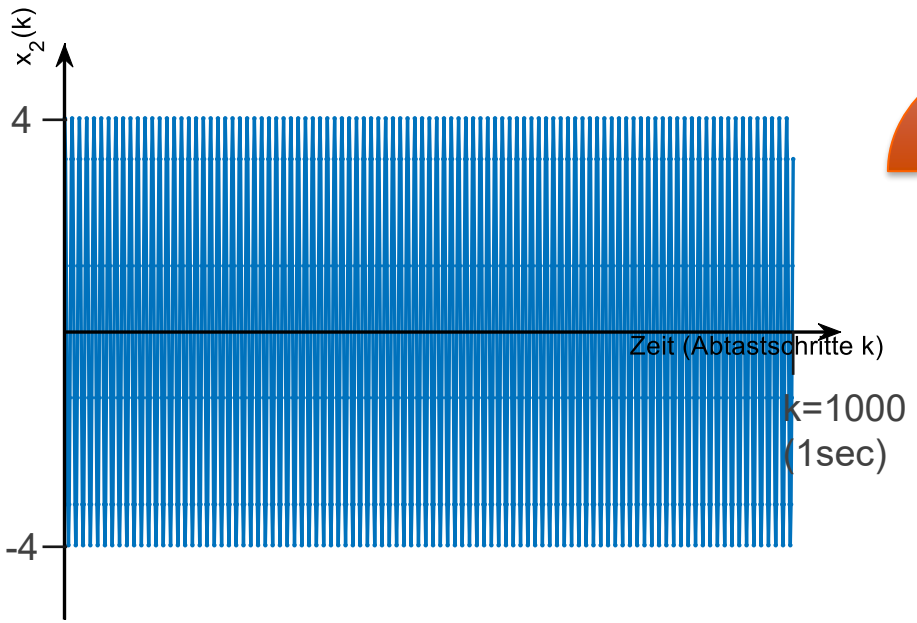
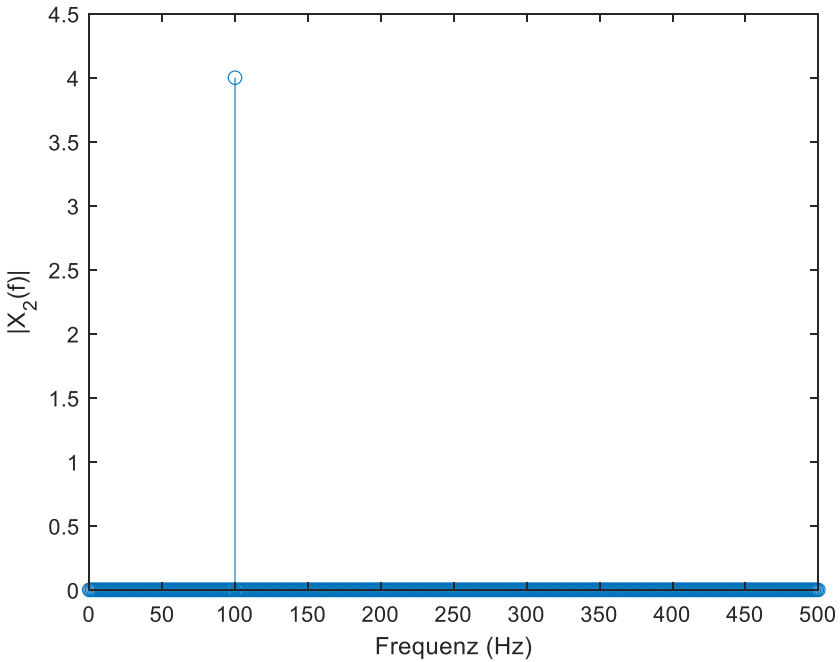


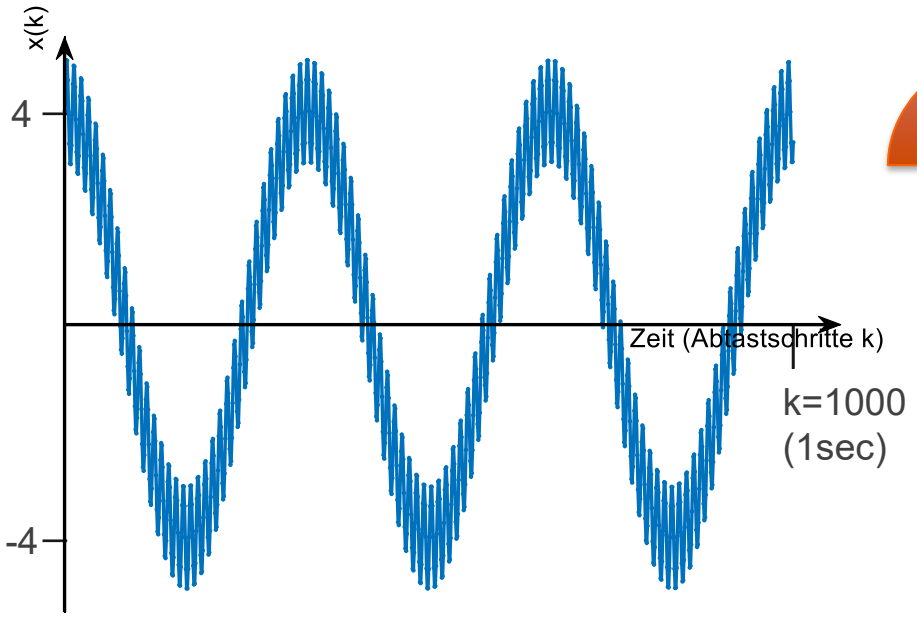



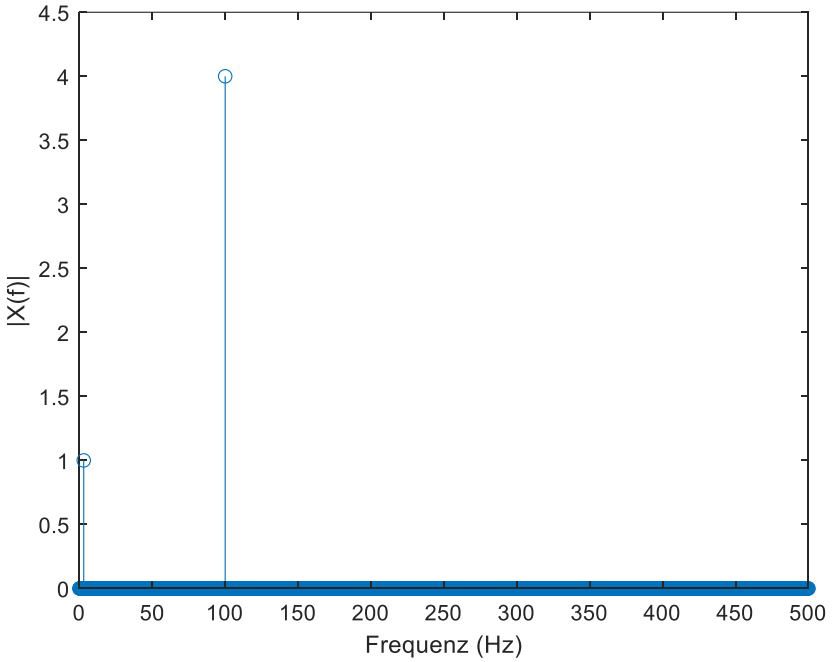

$$T_s = 1\text{ms}\,, F_s = \frac{1}{1\text{ms}} = 1\text{kHz}$$

$$x_2(\text{kT}) = 4\cos(2\pi F_2 kT)$$

$$F_2 = 100\text{Hz}$$

Fourier-Transformation
(DFT, FFT)



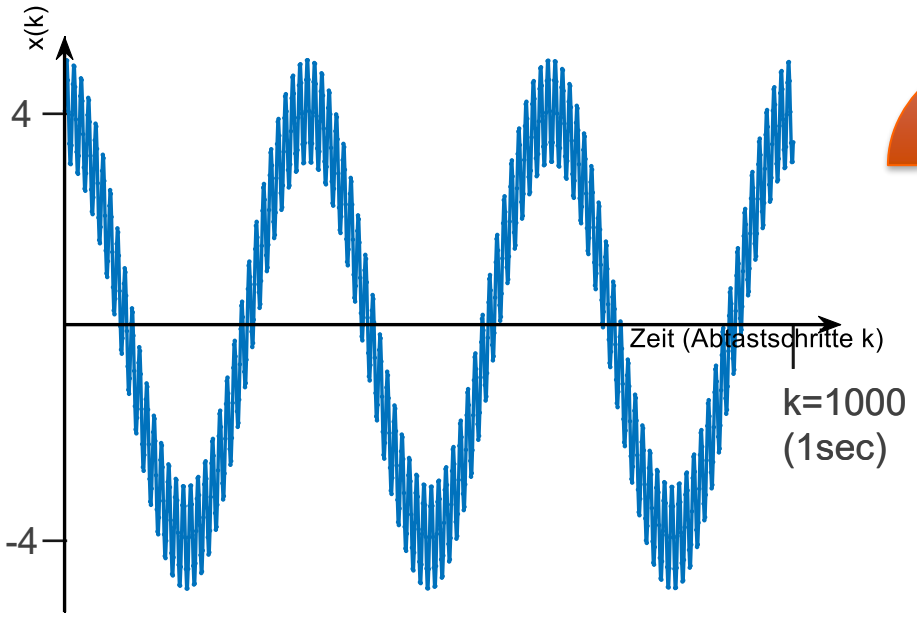



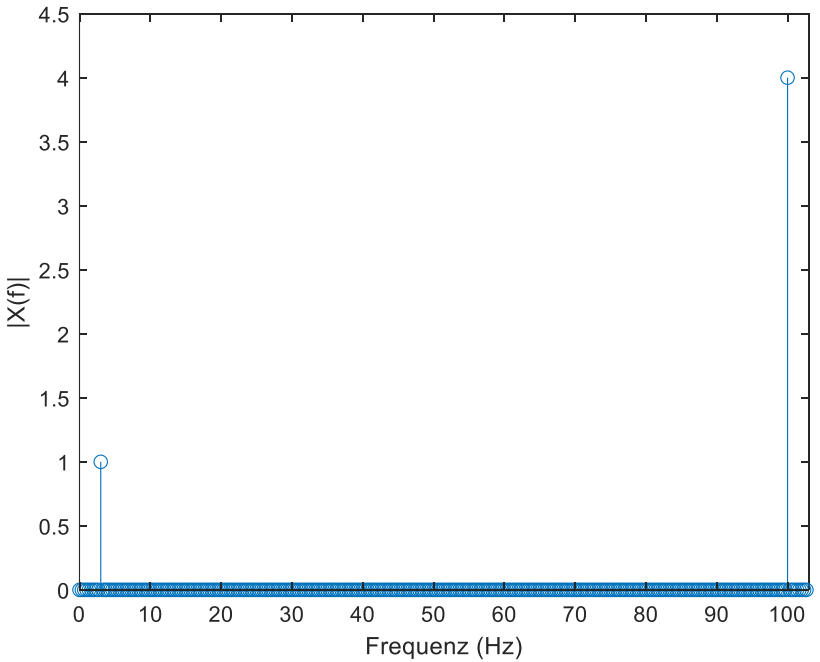

$$T_s = 1\text{ms}\,, F_s = \frac{1}{1\text{ms}} = 1\text{kHz}$$

$$x(\text{kT}) = x_1(\text{kT}) + x_2(\text{kT}) = \sin(2\pi F_1 kT) + 4\cos(2\pi F_2 kT)$$

$$F_1 = 3\text{Hz}, F_2 = 100\text{Hz}$$

Fourier-Transformation
(DFT, FFT)




$$T_s = 1\text{ms}\,, F_s = \frac{1}{1\text{ms}} = 1\text{kHz}$$

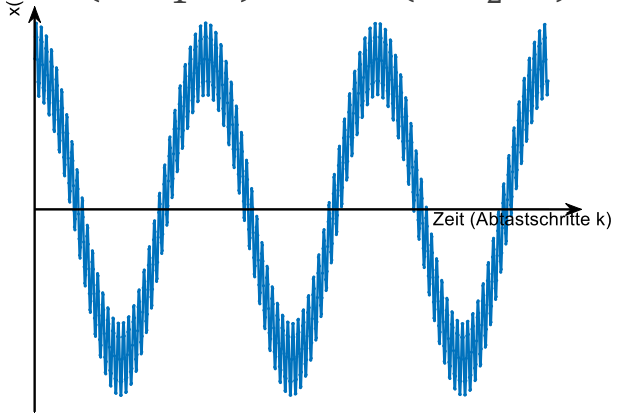
$$x(\text{kT}) = x_1(\text{kT}) + x_2(\text{kT}) = \sin(2\pi F_1 kT) + 4\cos(2\pi F_2 kT)$$

$$F_1 = 3\text{Hz}, F_2 = 100\text{Hz}$$

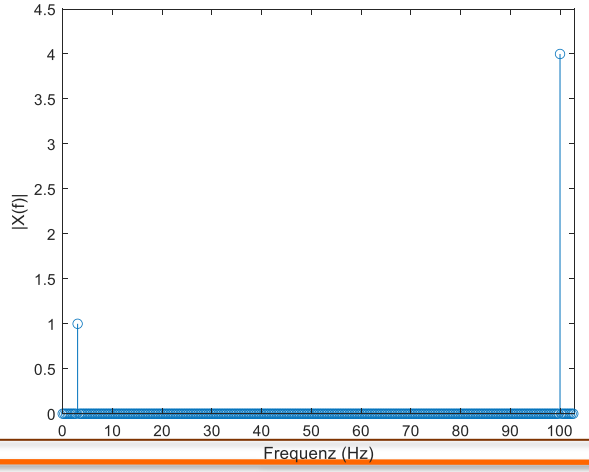
$x(\mathrm{kT})$ → **Digitalfilter** → $y(\mathrm{kT})$

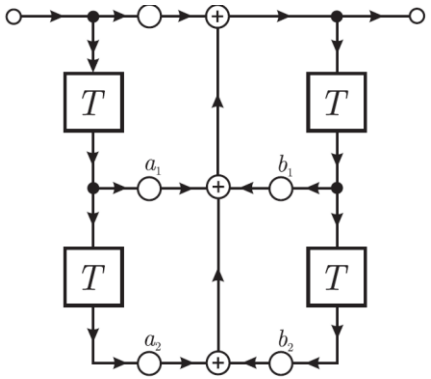
$\sin(2\pi F_1 kT) + 4\cos(2\pi F_2 kT)$

Zeitbereich



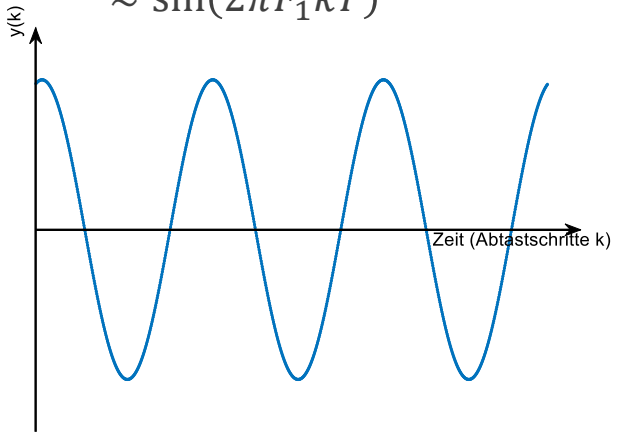

Frequenzbereich



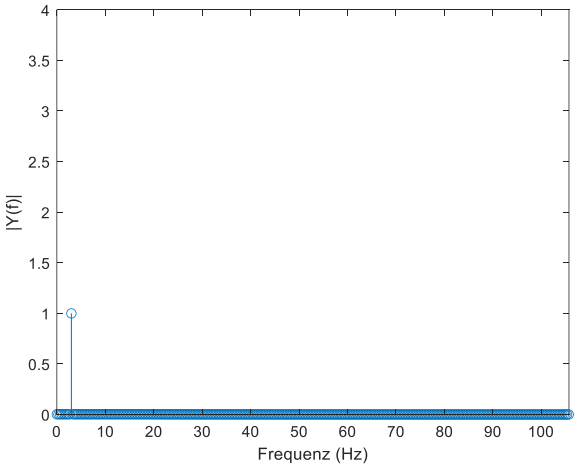



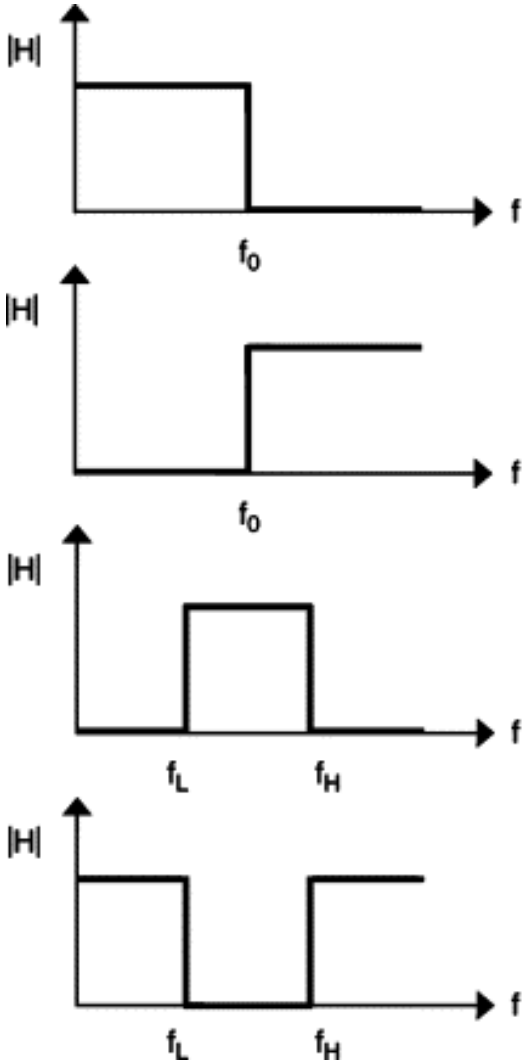

$\approx \sin(2\pi F_1 kT)$

Zeitbereich




Frequenzbereich






Ideal low-pass filter

Ideal high-pass filter

Ideal band-pass filter

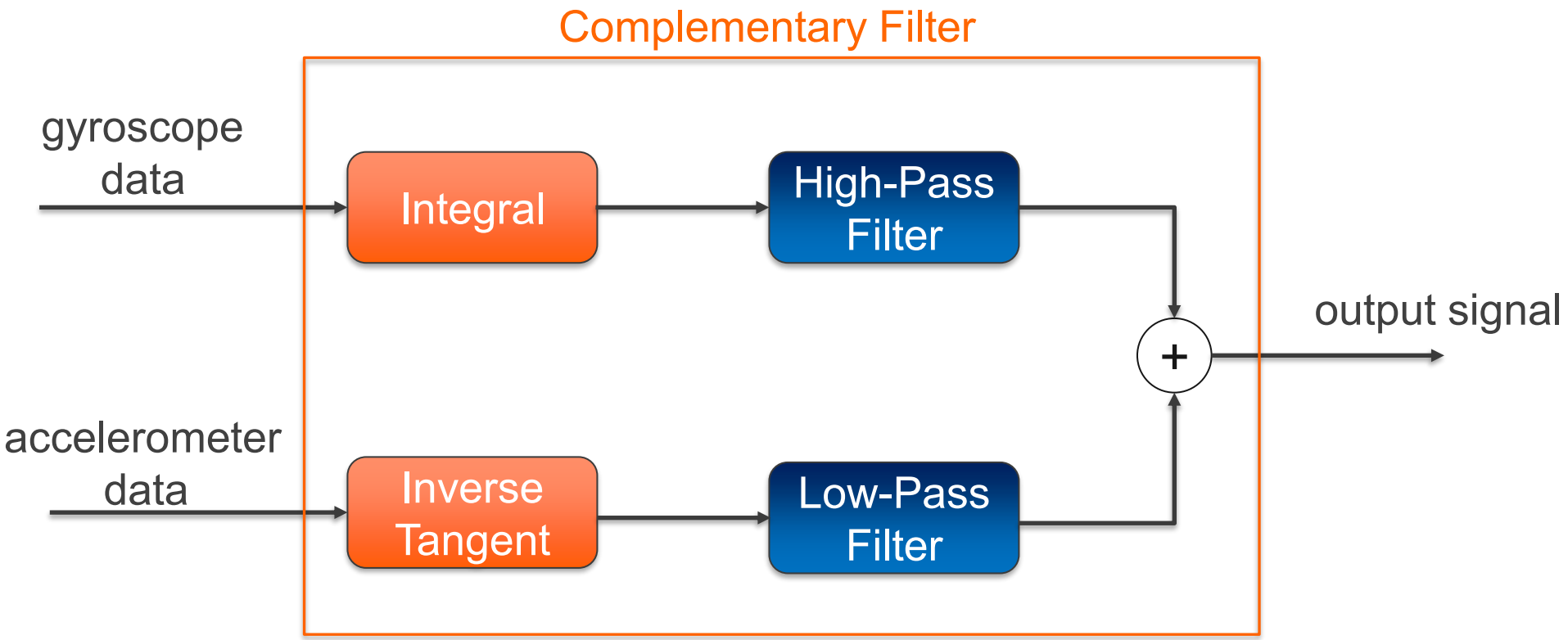
Ideal band-stop filter

## Common Use Cases for Digital Filters

- **Noise reduction:** Suppression of unwanted noise in sensor, audio, image, and video signals
- **Signal amplification / differentiation:** Emphasis of specific frequency components of a signal (e.g., to enhance ultrasound images)
- **Equalization:** User- or application-specific frequency response shaping (e.g., bass boost, speech focus)
- **Image processing:** Image sharpening, smoothing, and feature enhancement
- **Communication systems:** Channel selection, noise suppression in modulation/demodulation chains (radio, cellular, WLAN)
- …

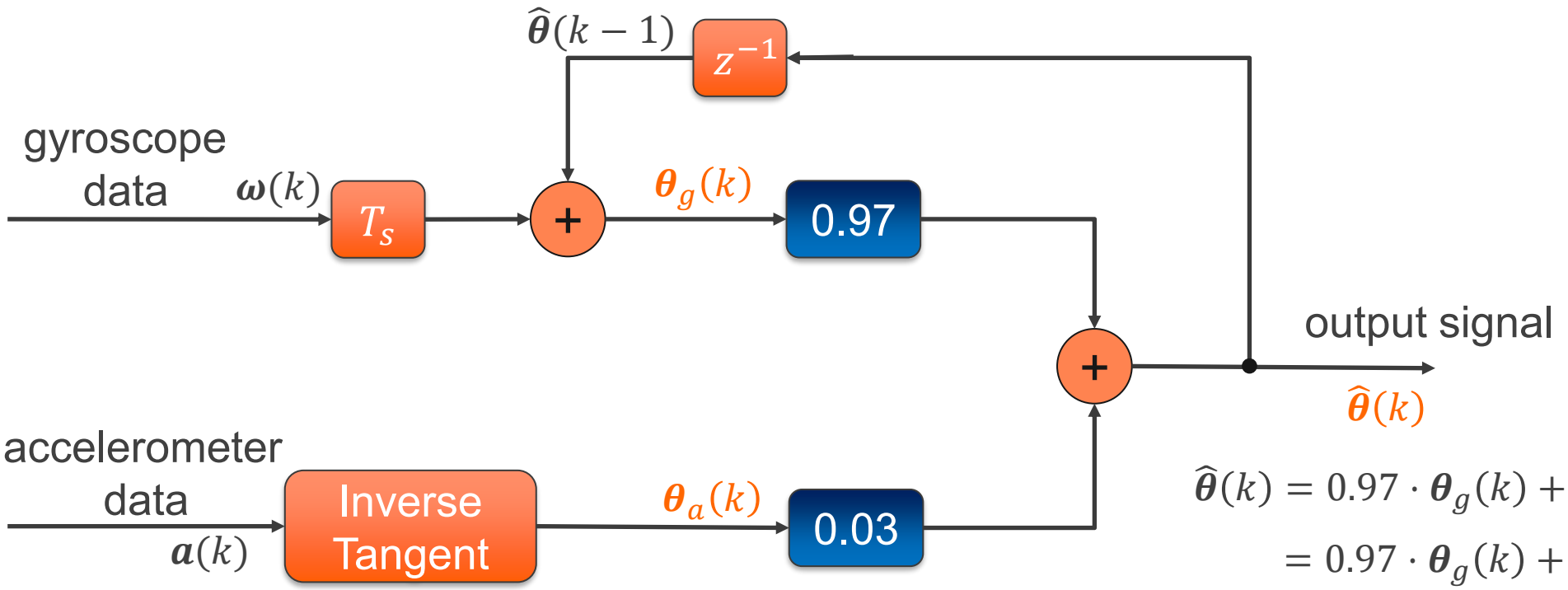
# Complementary Filter

Principle of complementary filter:




- **High-pass filter:** preserves the short-term, high-frequency orientation changes from the gyroscope while suppressing low-frequency drift caused by integration errors.
- **Low-pass filter:** preserves the long-term, low-frequency orientation information from the accelerometer while suppressing high-frequency noise and dynamic disturbances.

Discrete implementation of Complementary filter (for derivation see this video):




$$\hat{\boldsymbol{\theta}}(k) = 0.97 \cdot \boldsymbol{\theta}_g(k) + 0.03 \cdot \boldsymbol{\theta}_a(k)$$
$$= 0.97 \cdot \boldsymbol{\theta}_g(k) + (1 - 0.97) \cdot \boldsymbol{\theta}_a(k)$$
$$= \gamma \boldsymbol{\theta}_g(k) + (1 - \gamma) \cdot \boldsymbol{\theta}_a(k)$$

$$\boxed{\hat{\boldsymbol{\theta}}(k) = \gamma \boldsymbol{\theta}_g(k) + (1 - \gamma) \cdot \boldsymbol{\theta}_a(k)}$$

## Exercise L6.1 – Sensor Fusion using a complementary filter (MATLAB)

- Implement the complementary filter based on previous slide in MATLAB
- Use it for combining the accelerometer and the gyroscope data contained in ‘no_motion.mat’
- Start with a value of $\gamma = 0.97$
- Plot the different results into a single figure with 3 subplots:
  - 1. subplot: euler angles based on gyroscope data
  - 2. subplot: euler angles based on accelerometer data
  - 3. subplot: euler angles based on complementary filter (combining gyroscope and accelerometer data)

  Use the same axis scaling for subplots 2 and 3 for better comparison
- Play around with different values for $\gamma$ and check the effect
- Test the algorithm also with sensor_data_2.mat

## Exercise L6.2 – Sensor Fusion using a complementary filter (Simulink)

- Implement the complementary filter based in Simulink
- Download the file ex_6_2_template.zip, extract it and complete the missing pieces in the simulink model

Note: if you want to change the loaded mat-file or the value for gamma, change the corresponding values in the “load_data.m” which is part of the zip file and runs automatically when starting the Simulink model